

# DHRUV RAJPUT

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## EDUCATION

### Thapar Institute of Engineering and Technology

Patiala, Punjab, India

### Bachelor of Technology (B.Tech), Computer Science

Aug 2024 – June 2028

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Operating Systems, Cloud Computing, DevOps

## TECHNICAL SKILLS

Languages: Java, C++, C, Python, Go (Golang)

AI / ML: LangChain, LangGraph, RAG, OpenAI API, ChromaDB, Qdrant, Prompt Engineering

Web & APIs: FastAPI, React, TypeScript, REST APIs

Cloud (AWS): AWS, EC2, S3, IAM, Load Balancers

DevOps & Tools: Docker, CI/CD Pipelines, Git, GitHub

Core CS: Data Structures & Algorithms, OOP, Computer Networks, Operating Systems

## PROJECTS

### Text-to-SQL Query Engine | Python, FastAPI, React, OpenAI API, ChromaDB, LangChain

- Built an end-to-end RAG pipeline embedding database schemas into a ChromaDB vector store and retrieving only the top-3 relevant tables per query, eliminating context-overflow issues common in naive text-to-SQL systems
- Engineered a semantic layer annotating every column with business-context descriptions to guide the LLM toward accurate joins, aggregations, and filters
- Implemented a HITL safety guard blocking non-SELECT SQL (INSERT, UPDATE, DELETE, DROP) and requiring explicit user confirmation before execution
- Built a React + TypeScript frontend with syntax-highlighted SQL display, a pageable results table, and a schema explorer, connected to a FastAPI backend with observability logging

### Enterprise Agentic RAG Pipeline | Python, LangGraph, FastAPI, Qdrant, Portkey

- Built a production-grade agentic RAG system with LangGraph for cyclic reasoning, multi-step planning, and conversation memory, distinguishing true technical data from noisy inputs via semantic re-ranking
- Implemented NeMo Guardrails to block off-topic, jailbreak, and injection inputs before retrieval, and routed all LLM calls through a Portkey gateway with automatic fallback between primary and backup Groq keys
- Built an enterprise search layer using Qdrant Cloud vector search, 3072-dim Gemini embeddings, and FlashRank local reranking for high-performance retrieval
- Developed a RAGAS-powered evaluation suite (6 metrics) with full observability via Pydantic Logfire and LangSmith tracing across every agent node

### Electricity Prediction AI | Python, Machine Learning

- Built a machine learning model to predict electricity consumption patterns using historical time-series data with regression and time-series analysis techniques
- Implemented data preprocessing pipelines, feature engineering, and model evaluation metrics to improve forecasting accuracy
- Visualized prediction results and deployed the model for practical energy demand forecasting use cases

### Network Traffic Classification | Python, Cybersecurity, ML

- Designed and trained a machine learning model to classify network traffic for cybersecurity threat detection
- Analyzed packet-level and flow-level features to distinguish between normal and malicious traffic patterns
- Applied supervised learning algorithms and feature engineering, bridging Computer Networks and ML concepts to address real-world security challenges

## CERTIFICATIONS AND ACHIEVEMENTS

### AWS Agentic AI Demonstrated | AWS Training & Certification

Jul 2026

- Completed AWS's applied microcredential validating hands-on skills in designing, building, and deploying agentic AI workflows

### AWS Cloud Quest: Cloud Practitioner | AWS Training & Certification

Aug 2026

- Completed AWS's hands-on, gamified learning path covering core cloud concepts, services, and best practices

### AWS Academy Graduate – Cloud Foundations | AWS Academy

Aug 2026

- Completed a 20-hour AWS Academy training program covering core cloud computing concepts, architecture, and AWS foundational services

### Networking Basics | Cisco Networking Academy

Aug 2026

- Achieved student-level credential covering network communication, IPv4/IPv6 addressing, router configuration, and connectivity troubleshooting